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DUAL-MODE INSULATOR CLEANING: FROM AUTONOMOUS NAVIGATION TO HAPTIC TELEOPERATION USING INSULATOR 6D POSE

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ABSTRACT

Overhead power line insulators are exposed to environmental contaminants such as dust, salt, and pollutants, which can degrade their performance. Traditional cleaning methods often require manual labor and de-energizing power lines, leading to downtime and safety risks. This paper presents a dual-mode robotic system designed for efficient insulator cleaning. The system includes a 7-DoF robotic manipulator equipped with an RGB-D camera and a force/torque sensor for accurate perception and interaction. A YOLO-based vision module segments the insulator, followed by 6D pose estimation using structured point cloud data and known insulator dimensions. The control architecture combines autonomous alignment with operator-guided cleaning through a haptic interface that provides real-time force feedback. All modules are synchronized via a low-latency REmote DIctionary Server (Redis) network to ensure seamless operation. The estimated 6D pose was validated using an optical motion capture system, yielding a mean positional error of 3 cm along the z-axis and less than 1 cm on other axes. In 25 experimental trials, the system achieved an 88% success rate in autonomous alignment. These results highlight the system's precision, safety, and responsiveness—marking a step toward live-grid robotic maintenance.

Keywords: 6D Pose Estimation, Insulator Cleaning, Teleoperation, Haptic Feedback, Manipulation, Dual-mode Robotic system

NOMENCLATURE

6D Pose	Six-Dimensional Pose
DoF	Degrees of Freedom
F/T sensor	Force / Torque Sensor
HV	High Voltage
PCD	Point Cloud Data
Redis	REmote DIctionary Server
RGB-D	Red Green Blue with Depth
PCA	Principal Component Analysis
$\mathbf{I}$	RGB Image
$\mathbf{P}$	PCD
h	height of the insulator
r	radius of the insulator
$\mathbf{M}$	Binary Mask
$\mathbf{R}$	References points on image
$\mathbf{P_i}$	PCD of insulator
$\mathbf{V}$	Valid points on insulator
L	Line Equation
$\mathbf{P_s}$	Pose of insulator
$\mathbf{T}$	Transformation matrix

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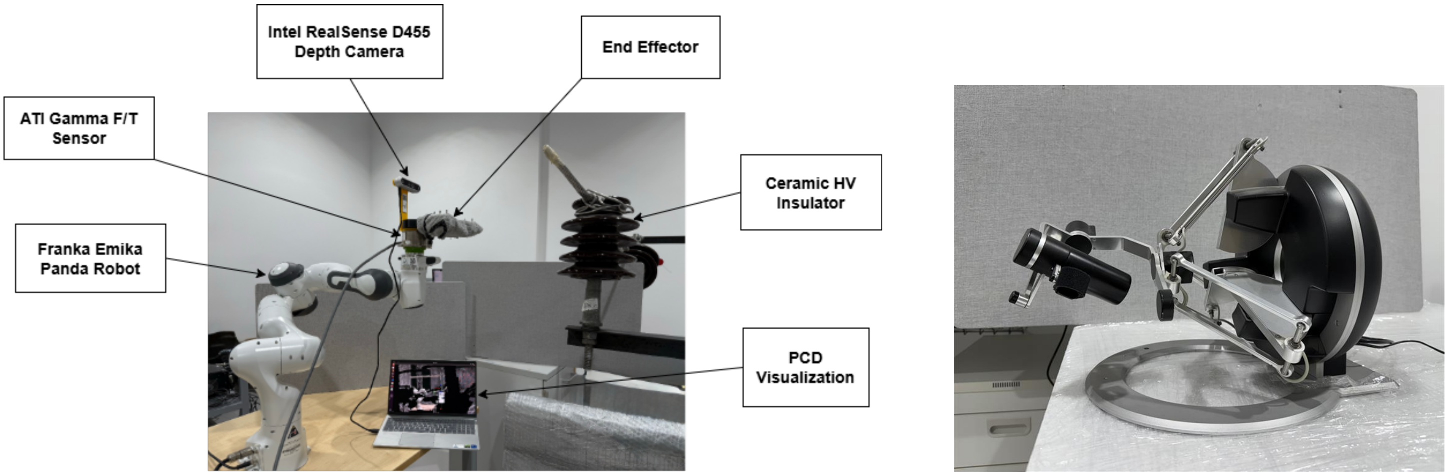
1. INTRODUCTION

The maintenance of overhead power line insulators is critical to ensuring the reliability and safety of electrical power transmission. These insulators accumulate contaminants such as dust, salt, and industrial pollutants, which, when combined with moisture, can lead to leakage currents, flashovers, and costly outages [1, 2]. Traditional cleaning methods typically require manual labor and de-energizing power lines, resulting in extended downtime and safety risks [3, 4]. These challenges have driven significant interest in robotic solutions to improve efficiency, safety, and enable live-line operations.

1.1. Robotic Systems for Power Line Maintenance and Cleaning

Robotic interventions for power line maintenance have evolved over decades, initially focusing on teleoperated systems that remove humans from hazardous environments. For example, Aracil et al. (2002) presented the ROBTET system for live-line maintenance up to 90 kV, highlighting the need for specialized vehicles and operator supervision [5]. Similarly, Kyushu Electric Power Co. (KEPCO) developed hot-line maintenance robots that progressed from bucket-operated manipulators to semi-automatic



(a) Manipulator with camera

(b) Omega.7 haptic device

FIGURE 1: Setup for dual-mode robotic system

systems with automatic tool changers and visual aids [6]. These early efforts demonstrated robotics' potential to enhance safety and task consistency. More recently, Banthia et al. (2018) introduced a telerobotic platform with hydraulic manipulators and haptic feedback for dexterous remote operation [3]. Other works target substation insulator cleaning [7], bio-inspired robots with inchworm locomotion and dry brushing [4], and autonomous robots for challenging environments [8]. Chen et al. (2024) developed a robotic arm for porcelain bushing rinsing using 5G communication [9], while drone-robot hybrids for insulator chain cleaning illustrate both the advantages and limitations of aerial platforms [10].

1.2. Autonomous Navigation and Operation

Increasing autonomy in power line robotics aims to reduce operator workload and improve operational efficiency. Feng and Zhang proposed an autonomous live-line maintenance robot (ALMR) for 10 kV lines, featuring hierarchical task planning and cable recognition in complex outdoor settings [11]. UAV-based systems are advancing as well; for example, Zeng et al. (2023) addressed autonomous landing of hybrid flying-walking inspection robots using prior structure data and real-time semantic segmentation [12], and Zou et al. (2023) simulated a single-arm UAV-mounted inspection robot focusing on mechanical design and dynamics [13].

1.3. 6D Pose Estimation

Accurate 6D pose estimation (3D position and orientation) is essential for robotic interaction with insulators, especially for precise cleaning tool alignment. Earlier approaches such as Drost et al. (2010) employed oriented point pair features in point clouds for robust object recognition [14]. Deep learning methods using RGB or RGB-D data have significantly advanced the field. PoseCNN estimates object translation and rotation using RGB images with a novel loss for symmetric objects [15]. Kehl et al. (2017) introduced SSD-6D for fast 3D object detection from RGB images trained on synthetic data [16]. Zakharov et al. (2019) proposed DPOD for dense 2D-3D correspondences and pose refinement [17], while Pix2Pose [18] predicts 3D coordinates per pixel without requiring textured models. Labbé et al. (2020) developed CosyPose for multi-object 6D pose recovery with scene-level refinement [19].

1.4. Haptic Teleoperation

While autonomy is desirable, teleoperation with haptic feedback remains critical for handling complex tasks and real-time uncertainties. Banthia et al. (2018) investigated force feedback mechanisms to enhance operator performance during live-line maintenance despite disturbances and delays [3]. The ROBTET system also employed master arms with force feedback [5]. Dual-arm coordination in mobile power cable robots has been studied with a focus on minimizing internal forces under various operational modes [20].

1.5. Problem Statement

Robotic Maintenance of overhead power line insulators requires both accurate positioning of the cleaning tool and adaptive contact-based cleaning. For effective cleaning, the end-effector must be precisely aligned with the target insulator surface. Fully autonomous systems, while efficient, can pose a risk of damage due to estimation of errors or false detections during contact tasks. Conversely, fully manual teleoperation increases operator workload and reduces system efficiency. To address these limitations, this work presents a dual-mode robotic system that integrates autonomous 6D pose estimation of insulators using RGB-D data with real-time haptic teleoperation. The system autonomously aligns the cleaning tool with the insulator and then seamlessly transitions control to the operator for precise cleaning using tactile feedback. This integrated approach leverages the strengths of both autonomy and human-in-the-loop control—ensuring accurate alignment, preserving operator dexterity for fine manipulation, reducing task fatigue, and enabling safe, efficient live-line maintenance without power shutdown.

Building on these foundations, this paper presents a dual-mode robotic system (Fig. 1a) featuring a haptic controller (Fig. 1b) specifically designed for insulator cleaning. The system combines autonomous 6D pose estimation and alignment using RGB-D data with real-time haptic teleoperation to enable precise cleaning. Extensive experiments demonstrate the effectiveness of the proposed system. The key contributions of this work are:

- A 6D pose estimation method for insulators, validated via optical motion capture.
- A dual-mode control system combining autonomous alignment and haptic-guided cleaning.
- Real-time integration of perception and control over a low-latency Redis network.
- Experimental validation on a lab-scale insulator setup.

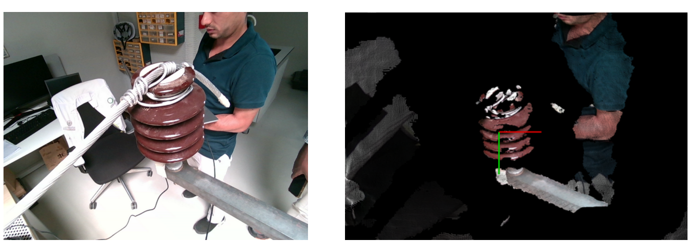
The rest of the paper is organized as follows: Section 2 details the 6D pose estimation method; Section 3 describes the robotic system, hardware integration, and workflow; Section 4 concludes and discusses future work.

2. PROPOSED METHOD FOR 6D POSE ESTIMATION

This section presents our method for estimating the 6D pose of the insulator by utilizing its known physical dimensions—height (h) and radius (r). The 6D pose refers to the object's three-dimensional position (x, y, z) and orientation (roll, pitch, yaw) relative to a reference frame, which in our case is the camera frame. Our approach combines semantic segmentation on RGB image data to identify the insulator and 3D geometric operations on the corresponding point cloud data from an RGB-D camera to estimate its pose. We describe each stage of the processing pipeline in detail and conclude with experimental validation results demonstrating the method's accuracy and effectiveness.

2.1. Data Acquisition System

The data acquisition process utilizes an Intel RealSense Depth Camera D455, which simultaneously captures RGB images (Fig. 2a) and registered point cloud data (Fig. 2b) of the target insulator and its surroundings. Point cloud data consists of thousands of 3D points that represent the surfaces of objects in the scene. This raw data contains considerable noise and irrelevant environmental information that must be filtered out to accurately determine the insulator's 6D pose.



(a) 2D RGB image **(b) Registered PCD**

FIGURE 2: Data from depth camera

2.2. Image Processing

The initial processing step involves identifying and segmenting the insulator from the RGB image, represented as **I**, using YOLOv8 (You Only Look Once version 8) learning model trained on a custom dataset of insulator images. All processing was performed on a Dell laptop equipped with an Intel i7 processor, 16GB RAM, NVIDIA GeForce RTX 3050 GPU, and running Linux Ubuntu 20.4. As shown in Table 1, the image segmentation process takes an average of 111 milliseconds for a 640 × 480 resolution image, striking a balance between accuracy and computational efficiency. This segmentation produces a binary mask, represented by **M**, containing only the insulator's contours (Fig. 3a), which serves as the foundation for subsequent analysis. A binary mask is a black and white image where white pixels represent the object of interest (the insulator) and black pixels represent the background.

TABLE 1: Average Time Taken for Each Step

Steps	$Avg.Time$ [ms]
Image Segmentation	111
Reference Point Selection	≈ 1
Pose Estimation	55
Total Time	167

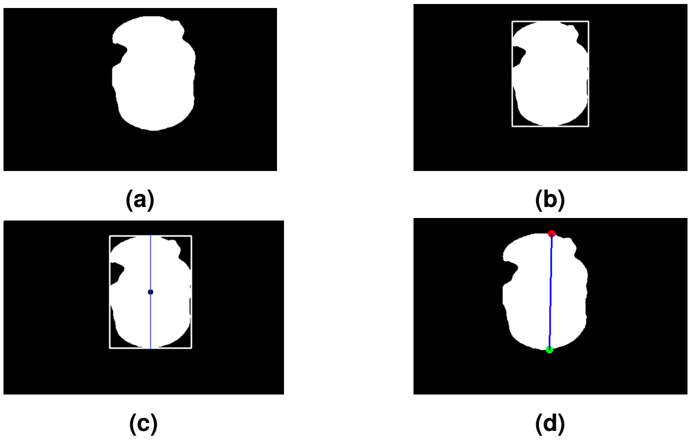
2.3. Central Axis Identification

Using the previously generated binary mask (**M**), we identify the central axis of the insulator through the following detailed process:

Establish a Minimum-Area Bounding Rectangle. We first identify the largest contour on the binary mask, which represents the boundary of the detected insulator shape. Then on the largest contour, using OpenCV's `cv2.minAreaRect()` function, we create a minimum area bounding rectangle, the smallest possible rectangle that can enclose the insulator contour regardless of its orientation. This rotated rectangle better fits elongated objects like insulators compared to axis-aligned bounding boxes (Fig. 3b).

Split the Bounding Rectangle Along Its Length. We then split this bounding rectangle through the middle along its length dimension (longer side) rather than its breadth. This approach provides a more accurate approximation of the insulator's functional axis and remains robust to variations in insulator orientation. The resulting line segment passes through the center of the bounding rectangle and effectively approximates the central axis of the insulator. (Fig. 3c).

Select Reference Points Along the Central Axis. Finally, we selected four strategically positioned points along the identified central axis, specifically from the top half, due to its greater visibility from the downward-facing camera. These points, which we refer to as "reference endpoints", represented by **R**, serve as spatial landmarks for subsequent analysis. We place particular emphasis on the two extreme points at the ends of the axis, which typically correspond to the terminal points of the insulator. These reference points are then projected from the binary mask coordinates back onto the original image for visualization and further processing. (Fig. 3d).



(a) (b) (c) (d)

FIGURE 3: Process for determining insulator orientation via central axis: (a) Binary mask. (b) Minimum bounding rectangle. (c) Division of the bounding rectangle along its length. (d) Reference point selection along the resulting central axis.

2.4. Point Cloud Filtering and Processing

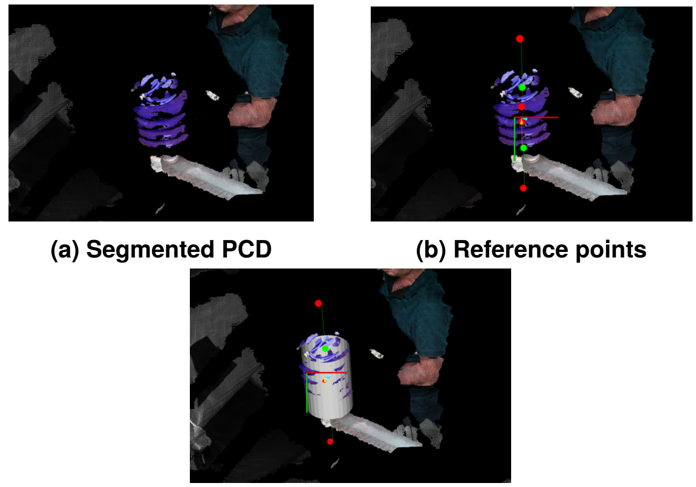
Concurrently, the binary object mask (**M**) is used to filter the PCD from camera, marked as **P**, isolating only those 3D points corresponding to the insulator, represented by $\mathbf{P_i}$ are shown in Fig. 4a. This step significantly reduces computational complexity and restricts the 6D pose estimation algorithm to the relevant spatial information. The four selected reference points are positioned along the insulator's central axis to capture the three-dimensional surface geometry from any viewing angle, with optimal spacing determined through empirical testing. Specifically, the top two reference points are positioned at approximately $2mm$ and $1mm$ from the center, deliberately placed closer together to compensate for perspective distortion in typical top-view camera angles. The lower two reference points are positioned at $20mm$ and $40mm$ from the center, with greater spacing to ensure comprehensive coverage of the insulator's lower section.

Each selected point must be examined for correspondence with actual point cloud data; when direct correspondences are unavailable, a nearest point search is performed using a KD-Tree algorithm. A K-Dimensional Tree (KD-Tree) is a space-partitioning data structure that organizes points in a k-dimensional space, allowing for efficient nearest neighbor searches. This verification ensures that only valid 3D points, represented by **V**, are utilized in subsequent calculations.

2.5. Axis Refinement with PCA

After obtaining the valid points **V**, a best-fit line (L) is computed using Principal Component Analysis (PCA) based on these endpoints. PCA is a statistical technique that analyzes the distribution of points in 3D space and identifies their primary direction of variation. This approach allows us to determine the directional alignment of the insulator more accurately than simple geometric methods, accounting for both pitch and roll angles of the insulator's orientation.

The fitted line is then visualized within the point cloud data, providing an approximation of both forward/backward and lateral tilt of the insulator. To better represent the insulator's longitudinal axis, we construct a parallel line offset from the surface-fitted line by a distance equal to the known radius (r, $73mm$) of the insulator. The offset direction is determined by the vector from the camera's optical center to the midpoint of the surface line (Fig. 4b), ensuring proper representation of the insulator's central structure.



(a) Segmented PCD **(b) Reference points**

(c) Cylinder Pose

FIGURE 4: Pose estimation steps

2.6. Final Pose Estimation and Implementation

After obtaining the refined central axis of the insulator, we proceed to determine its complete 6D pose. This step transforms our geometric understanding into a precise spatial representation suitable for robotic applications.

Cylinder Model Registration. For the final pose estimation, we employ the Point Cloud Library (PCL) due to its comprehensive suite of tools for 3D geometry processing. Using PCL, we:

1. Instantiate a parametric cylinder with dimensions matching standard insulators (diameter: 146mm × height: 210mm).

2. Position and orient this cylinder model to align with the previously extracted refined central axis (L).

As illustrated in Fig. 4c, this approach effectively aligns our parametric model with the actual insulator data, enabling accurate pose determination.

Mathematical Representation of 6D Pose. The registration process yields a transformation matrix **T** that encodes the complete 6D pose information of the insulator. This pose consists of:

- Position: 3D coordinates (x, y, z) representing the spatial location of the insulator's center

- Orientation: 3D rotation angles (roll, pitch, yaw) representing the insulator's alignment in space

The transformation matrix **T** is a 4×4 matrix that unifies both positional and orientational information in a single mathematical representation:

$$\mathbf{T} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where the 3×3 submatrix $[r_{ij}]$ represents the rotation component, and the vector $[t_x, t_y, t_z]^T$ represents the translation component. From this matrix, we extract the six parameters (3 for position, 3 for orientation) that constitute the complete 6D pose of the insulator.

Performance and Pipeline Integration. Our implementation achieves efficient performance, with the pose estimation phase requiring an average of 55 milliseconds as detailed in Table 1. The complete pipeline, from initial image acquisition to final pose output, is executed in just 167 milliseconds, enabling real-time applications.

Algorithm 1 provides a complete summary of our 6D pose estimation approach, while Fig. 5 visualizes the entire process from input data to final pose estimation. This efficient implementation ensures that our system can meet the demands of robotic applications while maintaining sufficient accuracy for insulator manipulation tasks.

Algorithm 1 6D Pose Estimation

Require: RGB image **I**
Require: Point Cloud **P**
Require: Insulator dimensions h,r

$\text{YOLOv8}(\mathbf{I}) \rightarrow \mathbf{M}$	▷ Get Binary Mask, section 2.2
$\mathbf{M} \rightarrow \mathbf{R}$	▷ Get the Reference Points, section 2.3
$(\mathbf{P}, \mathbf{M}) \rightarrow \mathbf{P_i}$	▷ Get the PCD for Insulator, section 2.4
$(\mathbf{P_i}, \mathbf{R}) \rightarrow \mathbf{V}$	▷ Get Valid 3D Endpoints, section 2.4
$(\mathbf{V}, PCA) \rightarrow \mathbf{L}$	▷ Get Line from Endpoints, section 2.5
$(L, \text{h}, \text{r}) \rightarrow \mathbf{P_s}$	▷ Get Insulator Pose, section 2.6

2.7. Performance Analysis

To evaluate the accuracy of the proposed 6D pose estimation method, we conducted a series of 25 experimental trials under varying initial viewpoints between the RGB-D camera and the target insulator. Ground truth poses were acquired using a Qualisys motion capture system, with reflective markers affixed to both the camera and the insulator to provide high-precision spatial tracking.

A detailed quantitative comparison was performed between the pose estimates generated by our system and the corresponding ground truth values. One representative example from Test 1 is presented in Table 2, where the estimated position is compared directly against the Qualisys reference.

Although the numerical difference in the Z-axis is approximately 29.8 mm, the practical impact of this deviation was negligible. Across all 25 trials, the robotic end effector was consistently aligned accurately with the insulator, enabling the successful execution of the task without manual adjustment. This

TABLE 2: Comparison of Qualisys ground truth and our estimated position for Test 1. All values are in millimeters.

System	X (mm)	Y (mm)	Z (mm)
Qualisys (Ground Truth)	2836.271	1113.843	1410.138
Our Estimation	2839.627	1113.505	1439.964
Difference	**-3.356**	**0.338**	**-29.826**

operational success, despite measurable position offsets, suggests that the discrepancy originates primarily from differences in coordinate system definitions between the Qualisys setup and the internal reference frame of the RGB-D camera.

Thus, while the system exhibits an average absolute positional error of up to 30 mm in some cases, the relative spatial alignment, critical for robotic manipulation, remains highly accurate. These findings confirm that the proposed method provides sufficiently precise pose estimates for reliable and repeatable robotic operation in real-world conditions.

3. ROBOTIC SYSTEM SETUP

Building on the 6D pose estimation pipeline outlined in Section 2, this section details the robotic hardware and software architecture developed to execute safe and precise insulator cleaning.

3.1. System Architecture Overview

The robotic system consists of a 7-DoF Franka Emika Panda manipulator, equipped with an Intel RealSense D455 RGB-D camera and a custom end-effector integrated with a six-axis force/torque sensor. It operates in dual-mode: first, it autonomously estimates the insulator pose and aligns the robot; then, it transitions to operator-guided teleoperation for the cleaning task, utilizing a haptic interface (Fig. 6). This design ensures precise positioning and force-controlled interaction during the cleaning of insulators.

3.2. Hardware Configuration

- **Manipulator:** The Franka Emika Panda arm provides 7 degrees of freedom, a 3 kg payload capacity, 850 mm reach, and 0.1 mm repeatability. Its torque-controlled joints enable compliant manipulation tasks involving contact.
- **End-Effector:** The current prototype features a microfiber cleaning pad for surface wiping, mounted on a modular end-effector designed for easy integration of task-specific tools in future iterations (Fig. 7).
- **Force/Torque Sensor:** An ATI Gamma F/T sensor mounted at the wrist provides six-axis force and torque feedback, ensuring safe manipulation and preventing damage to the insulator.
- **Perception Module:** An Intel RealSense D455 RGB-D camera is rigidly mounted on the end-effector. It streams aligned depth and RGB data at a resolution of 640 × 480 at 30 Hz. The static camera-to-end-effector transform simplifies extrinsic calibration and thus transforming pose from camera frame to robot frame (Fig. 7).

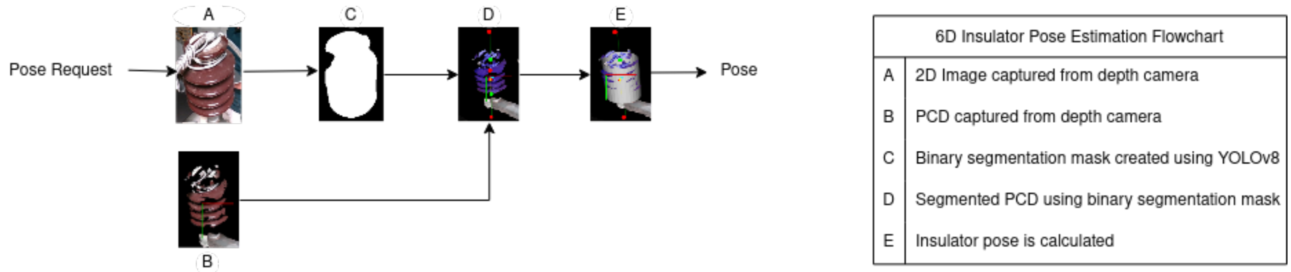


FIGURE 5: 6D Insulator pose estimation complete overview

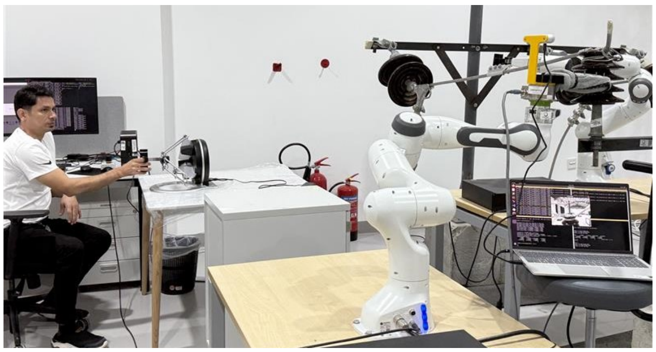


FIGURE 6: Complete workspace

- **Haptic Interface:** An Omega.7 haptic device (Force Dimension) is used for high-fidelity teleoperation. It offers a workspace of approximately 160 mm × 110 mm, sub-millimeter translational resolution (<0.01 mm), and angular resolution of 0.09°, enabling fine-grained operator control (Fig. 1b).
- **Communication Layer:** A Redis Network which supports low-latency and real-time data exchange between the manipulator, perception module, force sensor, and haptic device, is used.

(a) Front view **(b) Side view** **(c) Components**

FIGURE 7: End-effector and perception module

3.3. Workspace Configuration

The workspace includes a laboratory setup simulating an overhead transmission line structure with ceramic insulators. The setup operates in a non-energized mode, as the current focus is on developing and validating the core perception and control algorithms. At this stage, all experiments are conducted in a controlled laboratory environment. However, the system is intended to be progressively tested in outdoor scenarios to assess its robustness against real-world backgrounds, variable lighting conditions, and environmental factors. Future iterations will also explore deployment under high-voltage conditions to evaluate full-scale field readiness.

3.4. Dual-Mode Operational Workflow

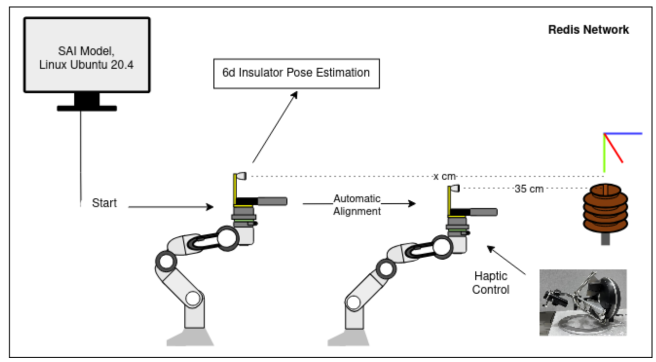


FIGURE 8: Operational workflow

Due to the limited workspace of the haptic device (160 mm × 110 mm), direct teleoperation is not feasible for full navigation, especially when the robot is distant from the insulator. To address this, the system incorporates an autonomous alignment mode. The workflow begins with the operator activating the vision module, which detects the insulator and estimates its 6D pose. The manipulator then autonomously aligns the end-effector with the insulator, stopping at a safe distance of 35 cm. Upon successful alignment, control shifts to the operator, who uses the haptic interface to guide the cleaning task, assisted by real-time force feedback from the force/torque sensor. The system ensures synchronized communication across all components through a low-latency Redis network, as shown in Fig. 8.

3.5. Observations

We evaluated the system's alignment accuracy over 25 trials. Each trial began by placing the robotic arm in a random configuration using haptic teleoperation, ensuring the insulator remained fully visible within the camera frame and positioned at a distance greater than the physical reach of the haptic device. Once the initial configuration was set, the vision module was triggered to estimate the insulator's pose. The robot then attempted to autonomously align its end-effector with the insulator. In 22 out of 25 trials, successful alignment was achieved, yielding an alignment accuracy of 88%. In the remaining cases, abrupt joint movements were observed due to complex inverse kinematics, triggering the system's safety halt mechanism.

4. CONCLUSION

This paper presented the development and validation of a dual-operation robotic system for insulator cleaning, combining autonomous alignment and operator-supervised teleoperation. The proposed system leverages RGB-D data for 6D pose estimation using semantic segmentation and 3D geometric processing, enabling precise localization of insulators. A Redis-based architecture ensures low-latency communication among perception, planning, and control modules. In 25 experimental trials, the system achieved an 88% success rate in autonomous alignment. For 6D pose estimation, validation against Qualisys motion capture data revealed a mean positional error of 3 cm along the z-axis and less than 1 cm along the remaining axes. These results confirm the system's effectiveness and precision. Future work will focus on extending testing to outdoor environments, including real and live-line insulator cleaning scenarios.

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REFERENCES

[1] Salem, Ali A. and Abd-Rahman, R. "A Review of the Dynamic Modelling of Pollution Flashover on High Voltage Outdoor Insulators." *Journal of Physics: Conference Series*, Vol. 1049. 1: p. 012019. 2018. IOP Publishing. DOI 10.1088/1742-6596/1049/1/012019.

[2] Tian, W., Guo, Weikang, Hu, Shuai, Sun, Haiping, Liu, Peng, Pan, Ziheng and Fu, Chunhua. "Effect of Volatility of Live Line Washing Agents on Physicochemical Properties of Silicon Rubber for Composite Insulator." *Proceedings of the 16th IET International Conference on AC and DC Power Transmission (ACDC)*: pp. 1996–2000. 2020. IET, Online. DOI 10.1049/icp.2020.0299.

[3] Banthia, Vikram, Maddahi, Yaser, Zareinia, Kourosh, Liao, Stephen, Olson, Tim, Fung, Wai-Keung, Balakrishnan, Subramaniam and Sepehri, Nariman. "A Prototype Telerobotic Platform for Live Transmission Line Maintenance: Review of Design and Development." *Transactions of the Institute of Measurement and Control* Vol. 40 No. 11 (2018): pp. 3273–3292. DOI 10.1177/0142331217732000.

[4] Wang, Lin, Wang, H. G., Song, Y. F. and Pan, X. A. "Development of a Bio-Inspired Live-Line Cleaning Robot for Suspension Insulator Strings." *International Journal of Simulation: Systems, Science and Technology* Vol. 17 No. 48 (2016): pp. 24–31. DOI 10.5013/IJSSST.a.17.48.24.

[5] Aracil, Rafael, Pinto, Emanuel and Ferre, Manuel. "Robots for Live-Power Lines: Maintenance and Inspection Tasks." *Proceedings of the 15th IFAC World Congress*, Vol. 35. 1: pp. 13–18. 2002. IFAC, Barcelona, Spain. DOI 10.3182/20020721-6-es-1901.00814.

[6] Maruyama, Yoshinaga. "Robotic Applications for Hot-Line Maintenance." *Industrial Robot: An International Journal* Vol. 27 No. 5 (2000): pp. 357–365. DOI 10.1108/01439910010373009.

[7] Pu, Ziheng, Xiong, Yuyao, Wang, Haitao, Yan, Biwu, Wu, Tian, Zheng, Lei and Yin, Pengxiang. "Design and Construction of a New Insulator Detection Robot for Application in 500 kV Strings: Electric Field Analysis and Field Testing." *Electric Power Systems Research* Vol. 173 (2019): pp. 48–55. DOI 10.1016/j.epsr.2019.03.025.

[8] Park, Joon-Young, Cho, Byung-Hak, Byun, Seung-Hyun and Lee, Jae-Kyung. "Development of Cleaning Robot System for Live-Line Suspension Insulator Strings." *International Journal of Control, Automation and Systems* Vol. 7 No. 2 (2009): pp. 211–220. DOI 10.1007/s12555-009-0207-7.

[9] Chen, Hao, Han, Wei, Xu, Weilun, Tang, Zongyao, Chen, Yini, Xu, Peng and Ma, Zhaoxing. "Design of Robotic Arm for the Porcelain Bushing in Substation." *Scientific Reports* Vol. 14 No. 1 (2024): p. 7698. DOI 10.1038/s41598-024-58443-7.

[10] Gonçalves, Rogério S., De Oliveira, Murilo, Rocioli, Murilo, Souza, Frederico, Gallo, Carlos, Sudbrack, Daniel, Trautmann, Paulo, Clasen, Bruno and Homma, Rafael. "Drone–Robot to Clean Power Line Insulators." *Sensors* Vol. 23 No. 12 (2023): p. 5529. DOI 10.3390/s23125529.

[11] Eng, Jiabo and Zhang, Weijun. "Autonomous Live-Line Maintenance Robot for a 10 kV Overhead Line." *IEEE Access* Vol. 9 (2021): pp. 61819–61831. DOI 10.1109/ACCESS.2021.3071243.

[12] Zeng, Yujie, Qin, Xinyan, Li, Bo, Lei, Jin, Zhang, Jie, Wang, Yanqi and Feng, Tianming. "A Novel Autonomous Landing Method for Flying–Walking Power Line Inspection Robots Based on Prior Structure Data." *Applied Sciences* Vol. 13 No. 17 (2023): p. 9544. DOI 10.3390/app13179544.

[13] Zou, Dehua, Liu, Lanlan, Jiang, Zhipeng, Luo, Bangwei and Quan, Wusheng. "Design and Dynamic Simulation of Single-Arm Inspection Robot System for Transmission Lines." *Proceedings of the 2023 International Conference on Advanced Robotics and Mechatronics (ICARM)*: pp. 1210–1213. 2023. IEEE, Xiamen, China. DOI 10.1109/ICARM.2023.000194.

[14] Drost, Bertram, Ulrich, Markus, Navab, Nassir and Ilic, Slobodan. "Model Globally, Match Locally: Efficient and Robust 3D Object Recognition." *Proceedings of the 2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*: pp. 998–1005. 2010. San Francisco, CA, USA. DOI 10.1109/CVPR.2010.5540108.

[15] Xiang, Yu, Schmidt, Tanner, Narayanan, Venkatraman and Fox, Dieter. "PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes." *Proceedings of Robotics: Science and Systems (RSS)*. 2018. Pittsburgh, PA, USA. DOI 10.15607/RSS.2018.XIV.019.

[16] Kehl, Wadim, Manhardt, Fabian, Tombari, Federico, Ilic, Slobodan and Navab, Nassir. "SSD-6D: Making RGB-Based 3D Detection and 6D Pose Estimation Great Again."

Proceedings of the IEEE International Conference on Computer Vision (ICCV): pp. 1521–1529. 2017. Venice, Italy. DOI 10.1109/ICCV.2017.169.

[17] Zakharov, Sergey, Shugurov, Ivan and Ilic, Slobodan. "DPOD: 6D Pose Object Detector and Refiner." *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*: pp. 1941–1950. 2019. Seoul, South Korea. DOI 10.1109/ICCV.2019.00199.

[18] Park, Kihwan, Patten, Timothy and Vincze, Markus. "Pix2Pose: Pixel-wise Coordinate Regression of Objects for 6D Pose Estimation." *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*: pp. 7668–7677. 2019. Seoul, South Korea. DOI 10.1109/ICCV.2019.00775.

[19] Labbe, Yohann, Carpentier, Jean, Aubry, Michaël and Sivic, Josef. "CosyPose: Consistent Multi-View Multi-Object 6D Pose Estimation." *Proceedings of the European Conference on Computer Vision (ECCV)*: pp. 574–591. 2020. Glasgow, UK. DOI 10.1007/978-3-030-58545-7_34.

[20] Jiang, Wei, Yan, Yu, Yu, Lianqing, Peng, Menghuai, Li, Hongjun and Chen, Wei. "Research on Dual-Arm Coordination Motion Control Strategy for Power Cable Mobile Robot." *Transactions of the Institute of Measurement and Control* Vol. 41 No. 11 (2019): pp. 3235–3247. DOI 10.1177/0142331219853540.